

3.2 Polarization and network size

Our first question is whether the polarization depends on the network size. That is if polarization increases as more users enter the conversation. We are interested on the evolution of polarization across the studied period and for this we divide the data in days, obtaining a network with the Twitter discussion data for each day. To investigate if polarization is correlated with network features we plot the polarization as a function of network size and average degree in Fig. 5.

For Andalusia dataset, network size is correlated with polarization. Bigger networks have a higher polarization score than smaller networks, with a Pearson Correlation Coefficient (PCC) of 0.76 and a p-value of 6×10^{-24} . The correlation of the average degree of the network with the polarization is 0.92, with a p-value of 3×10^{-50} (see Table 3, where values in parentheses represent p-values, and statistically significant results ($p < 0.01$) are highlighted in bold).

For Barcelona, the correlation between the network size and polarization is 0.44 with a p-value of 1×10^{-5} . The average degree correlates with polarization with a PCC value of 0.43, with a p-value of 2×10^{-5} . Here, the largest networks are not the most polarised in the Barcelona dataset.

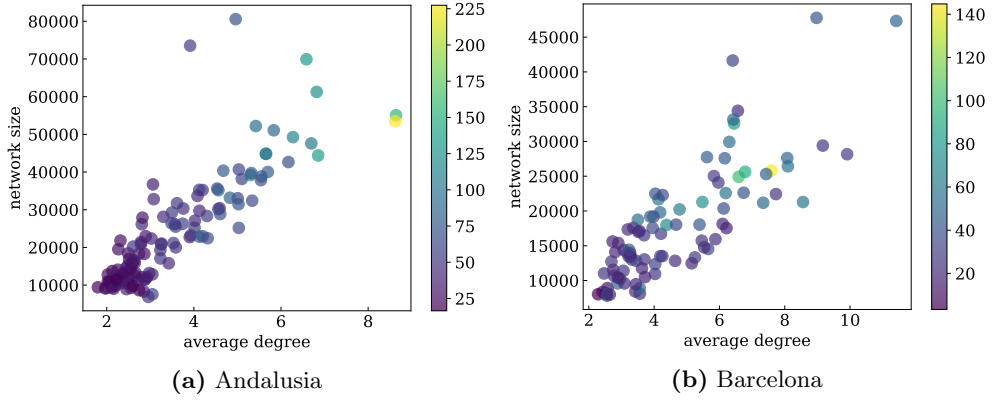


Fig. 5 The structural polarization for each day as a function of average total degree and network size.

Region	Network size	Average degree
Andalusia	0.7573 (6×10^{-24})	0.9187 (3×10^{-50})
Barcelona	0.4409 (1×10^{-5})	0.4334 (2×10^{-5})

Table 3 PCC of network size and average degree with polarization.

3.3 Polarization and hate speech

The presence of hate speech in our society poses a serious threat to harmonious coexistence between different social groups (Ezeibe, 2021). This type of rhetoric can become especially prominent during electoral campaigns or times of heightened political polarization, where it may be exploited for partisan gains (Romero-Rodríguez et al., 2023).

We examine the correlation between hate speech and polarization by calculating the percentage of tweets containing hate speech and the normalized polarization for each day across both datasets, as shown in Fig. 6.

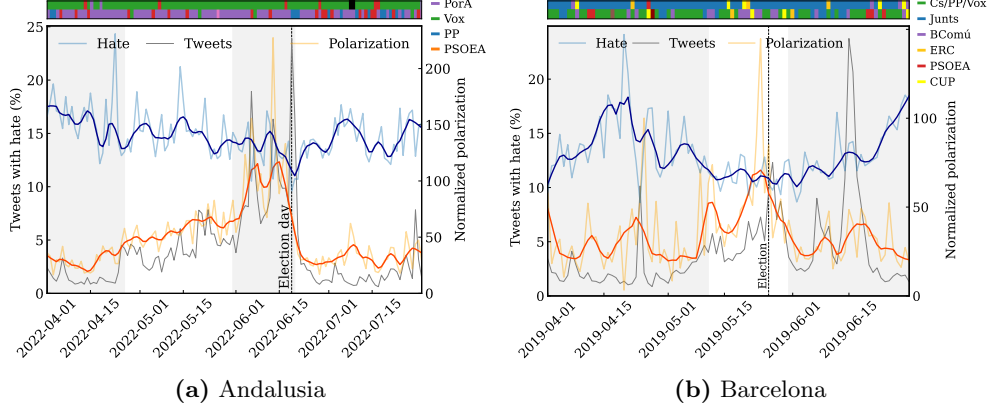


Fig. 6 Comparison between the percentage of tweets with hate speech and the polarization of the interaction network for each day. The thin grey line represents the volume of tweets per day. The grey shadow mark the change points boundaries. The dashed line indicates the election date. Smoothed using Locally Weighted Scatterplot Smoothing (LOWESS) with a window of 7 days. At the top there are the communities where the daily polarization poles belong.

For Andalusia, we can see that the polarization increases with the size of the network until the election day and abruptly decreasing after it. That confirms the expected impact of electoral campaigns polarising the political behaviours offline. The number of tweets containing hate speech also increases, but the number of tweets without hate speech rises even more, resulting in a decreasing percentage of tweets that contain hate speech. After the elections, we observe an increase in the percentage of hate speech, while the total number of tweets reaches a minimum and remains approximately constant. Before the elections, during the electoral campaign period, the polarization and hate speech are anti-correlated (see Table 4, where the same criteria as before for significance apply). There are two peaks of polarization occurring one and two weeks before the election, coinciding with the two days of televised debates between the candidates in the Andalusian 2022 regional elections that we are analysing (RTVE, 2022; CSRTV, 2022). Close to these two peaks of polarization, we also observe peaks in the number of tweets; however no peaks of hate are observed, only small fluctuations.

Region	Before elections	After elections
Andalusia	-0.4897 (4×10^{-6})	0.0106 (0.9)
Barcelona	-0.3805 (0.004)	-0.2509 (0.1)

Table 4 PCC of polarization and hate before and after elections.

In Barcelona, we observe that the percentage of hate speech is approximately constant throughout the studied period. However, there is a peak two months before the elections (April 24, 2019) coinciding with the date of the electoral debate for another election, the Spanish elections of April 28, 2019. This debate was broadcast on Catalan Public Television (TV3) and had a significant impact, becoming the most viewed general election debate in TV3’s history. It was also the most commented on television broadcast of the day on Twitter across Spain (TV3, 2019). Regarding polarization, we note that it reaches its highest value on the election date (June 26, 2019). Contrary to the case of Andalusia, polarization does not increase with the system size and fluctuates throughout all the entire period. Moreover, polarization increased after the elections, coinciding with the day of the investiture, which was marked by significant controversy. On this day, the number of tweets reaches its maximum, but not the level of polarization. On average, polarization in Andalusia is higher than in Barcelona, with an average value of 50 ± 3 compared to 35 ± 2 . Before the elections, there is anti-correlation between hate speech and polarization.

We compute the change point of the polarization and hate speech. For Andalusia, we identify three change points: the first occurs at the beginning of the Twitter discussion, the second one month before the election date, and the third the day of the elections. We then calculate the correlation between the percentage of hate speech and polarization for each period. However, these results are not statistically significant (see Table 5, where the same criteria apply). In the case of Barcelona dataset, we find two change points: one month before the election date and another just after the elections. Again, these results are not statistically significant.

Andalusia			Barcelona	
Period	End date	PCC	End Date	PCC
1	2022-04-26	-0.3464 (0.09)	2019-05-11	-0.2816 (0.08)
2	2022-05-31	-0.0700 (0.7)	2019-05-31	-0.0609 (0.8)
3	2022-06-20	-0.3862 (0.09)	2019-06-30	-0.0619 (0.7)
4	2022-07-31	0.0106 (0.9)		

Table 5 Breakpoints for timeseries of hate and polarization.

In Fig. 7, we present the detrended plot of polarization and hate speech, which results from removing the linear trend from the original time series data. This plot shows the fluctuations around the mean without the influence of the trend. The anti-correlation of polarization and hate speech is more pronounced for specific peaks